Baloney Detectives: The Misinformation Choose-Your-Own-Adventure Game

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ABSTRACT

For our final design project, our team prototyped a text-based game in Twine that aims to promote digital literacy in teenagers and help them fight against misinformation. It is meant to be a tool used by teachers as a digital literacy teaching aid. The game is an interactive way to engage teens in spotting mis- and dis-information. In our design, we have interviewed teenagers, who are direct stakeholders, to get their input on the game's concept and created "personas" of potential users to envision their experiences with our game.

KEYWORDS

Game design, misinformation, digital literacy, educational tools, social media

1 INTRODUCTION

Through *Baloney Detectives*, a video game created in Twine, we are addressing the convergent issues of misinformation and disinformation in social media. We are specifically addressing it by looking at the roles social media influencers play in spreading unintentionally or intentionally false information on the Internet. Our goal is to help teenagers learn techniques for identifying mis- and disinformation on social media. Our prototype is a way for them to be actively involved in “detective” work regarding health misinformation, something considered rampant in contemporary social media landscapes thanks to the ideologies spread by the brands of public figures.

According to a DataReportal survey, the 13-17 age range only makes up 6.6% of the Twitter user-base as of 2021 (Kemp). In 2022, Pew Research Center conducted a survey within the same age range and came up with 23% of teens saying they have ever used Twitter (Atske). It has been difficult to identify one number for what percentage of the userbase are teenagers due to surveys using different sampling methodologies and bin sizes. Even if only 6.6% of Twitter users are teens, the digital literacy concepts we are including in our game are applicable to other social media platforms. Twitter posts and influencers are also often shared through other social

[1]P03: Final Report
LIS-547: Design Methods for Librarianship
The Information School
University of Washington
Winter 2023

media or word of mouth. Digital literacy skills such as identifying fake posts and unreliable sources, however, are very valuable for evaluating information beyond Twitter itself.

2 Design Situation

Teens are our main stakeholders, as they are the targeted demographic, and their goal is to improve in information literacy and the detection of mis- or disinformation, which is a goal we are collaborating with them to achieve. Our project is also partially aimed at schools and libraries to purchase or adopt, as they may want to integrate the game into teen programming so that they can train young people in navigating information networks in the digital age and build skills in effective information literacy.

Teens are more susceptible to influencer messaging. According to the National Association of Anorexia Nervosa and Associated Disorders, 46% of 9-11 year-olds are either often or sometimes on diets and 35-57% of adolescent girls diet, fast, take diet pills, or engage in other potentially dangerous eating habits ("Eating Disorder Statistics"). We have thus identified a need to target misinformation surrounding harmful eating habits and promotion of diet products.

3 Design Process

A summary of the design process, design commitments, and goals. Best practice: Include a diagram of your process and include a timeline.
The body of your report will consist of a series of sections that give more details into your process and into your findings. These sections might be organized by design phase or by a sequence of methods.

Design Phase: Observation

1. Description of setting (context, physical, abstract flow models)

The Internet, chaotic as it is, has lots of potential for learning but also for harmful misinformation. Social media in particular allows for the quick, easy sharing of stories and information in short form, through the interface allowing for quick scrolling of massive amounts of content. It is easy to see what friends and colleagues are interested in based on what information is shared. Influencers on social media in general have many followers to whom they can promote and circulate ideas or advertise products, using their social presence as a platform, and users are more likely to trust posts shared by them due to that large following. Health/food trends and misinformation are especially easily spread through these sensationalized posts from influencers, often without being fact-checked.

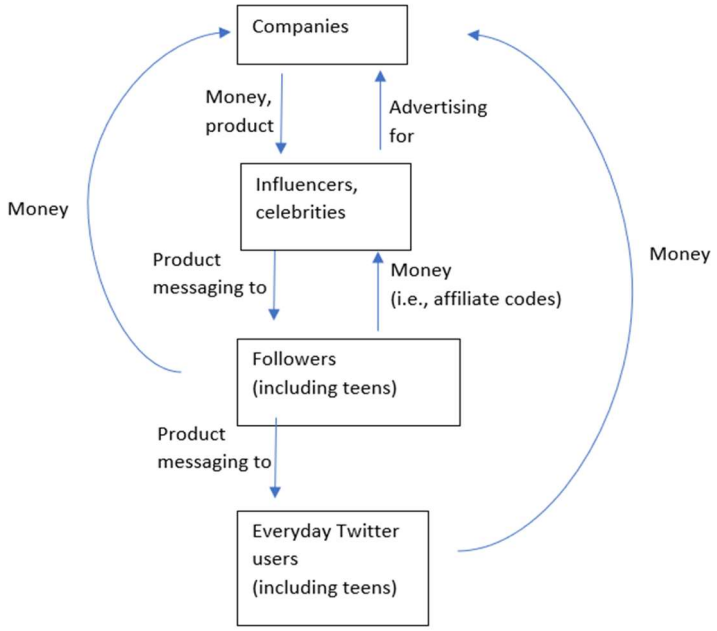


Figure 1. Stakeholder diagram of the misinformation problem we have identified

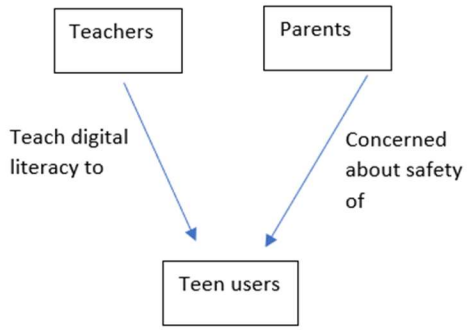


Figure 2. Stakeholder diagram for digital literacy solution

2. Results from semi-structured interviews or surveys

We luckily had two teens that were more than willing to work with us to develop the game further – Joe's teenage cousins Michael (age 15) and Alex (age 18), both of whom represent direct stakeholders at a variety of technology fluency and information literacy levels. Since they both enjoy video games and computers, we thought that it would be useful to gain an understanding of their specific needs through the interview.

We developed user personas for the cousins, based on the interviews, that helped inform how we could tailor the project to a variety of digital and information literacy levels. They are shown below, and we also created an affinity diagram based on their interviews.

Interview Notes and Results:

Michael:

- 18, Freshman at Carroll College in Helena, MT, current CS major.
- Knows basic social network analysis/understanding of how “liking” and “sharing”/reposting work through his coursework.
- VERY strong computer skills, including knowledge of social media.
- Made Twitter in high school but deleted it when Elon Musk took over. Presently uses Instagram and Discord significantly.
- Knows what misinformation means and could say an example, but would want some more meaningful support in understanding it.
- Does not know what “information literacy” means but wants to build skill when I explained it to him, because he thinks it is important.
    - “My college education hasn’t talked about this concept, so far”
    - No required coursework in information literacy at Carroll
- Gamer who hasn’t had time to play games due to college.
- Likes the idea of a game to combat fake news, thought title was funny.

Alex:

- 15, High School Sophomore in Bothell, more of a music/art kid but also likes STEM subjects (particularly CS/programming).
- Strong computer skills but is a fairly new social media user, mostly TikTok and Instagram but is aware of Twitter.
- Not aware of “misinformation” as a concept, but understood when I told him about fake news.
- Gamer, thinks that a game would be fun to help him understand what makes up misinformation and how to detect it.
    - Thinks it would be a good idea to use in school libraries or teen programming for public libraries when I explained it to him.
    - Would want a tutorial at first, but thinks he can get the hang of it over time.

3. Designer-generated scenarios

Scenarios 1:

David is a teenager who wants to become an influencer on social media. He plays Baloney Detectives and learns about the potential consequences of spreading misinformation and disinformation on social media platforms. He gains a better understanding of his responsibility as an influencer and the importance of verifying information before sharing it with his followers.

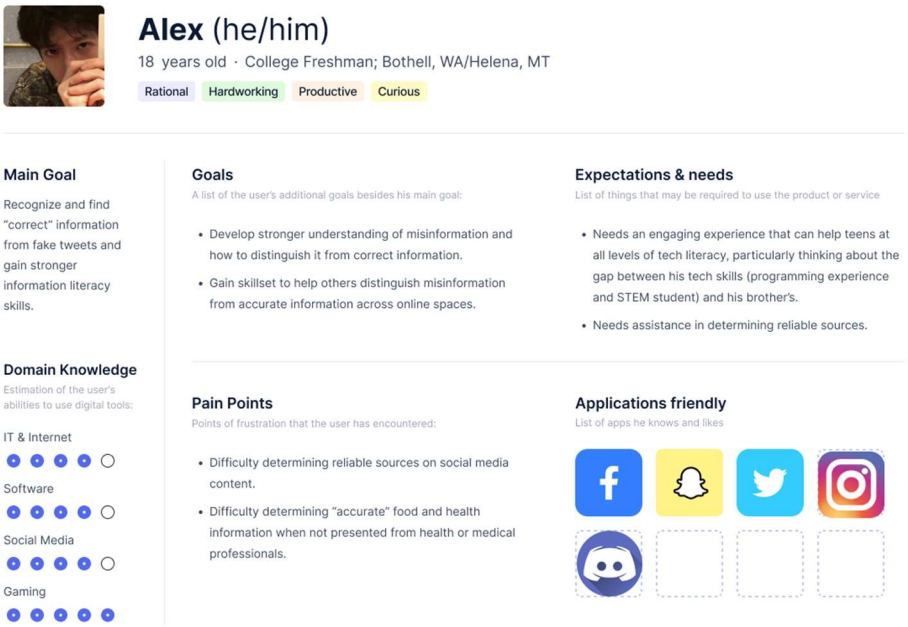
Scenarios 2:

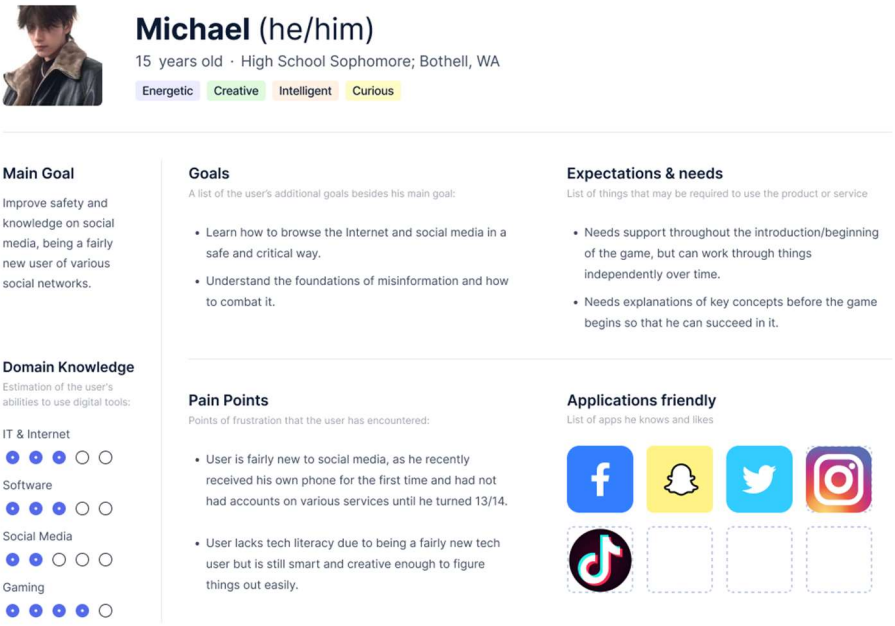
Emily is a social studies teacher who wants to engage her students in a discussion about the influence of social media on political campaigns. She uses Baloney Detectives as a tool to help her students understand how misinformation and disinformation can affect political discourse on social media. The students enjoy playing the game and learn valuable skills in media literacy, which they apply to their research on political campaigns.

Design Phase: Ideation

1. User Personas

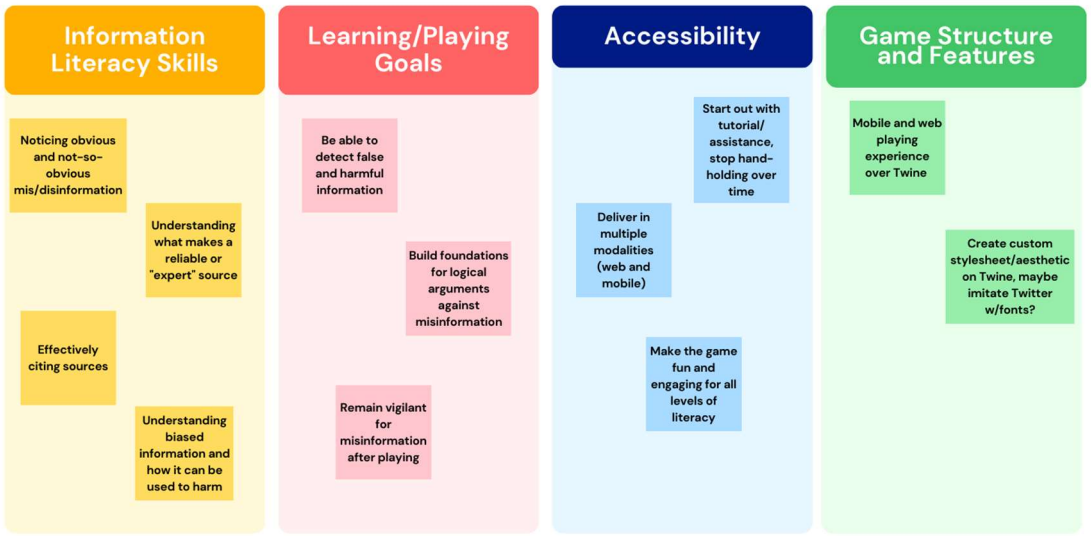
These user personas for Michael and Alex, based on the interviews, were created on the design software Figma and use insights gathered from the interviews to make suggestions on the prototypes and final product.





2. Affinity diagrams

This affinity diagram was made on the design software Canva and was based on insights gathered from the interviews with Joe's cousins, which suggested some features for the game and some learning and playing goals to utilize.



Design Phase: Prototyping

We have created a prototype. The prototype is a functional website that can be viewed on both web browsers and mobile phones. We have created a working prototype on Twine that is operational. It contains three different levels of mis/disinformation, which go from more obviously false to less obvious. The later two levels focus on evaluating sources. We are working on a final level that we will try to add in the next week. We are currently considering making this final level a synthesis of all concepts discussed. We would ask the users to combine all the skills gathered throughout the game as a "boss" level of sorts. However, the current prototype is operational and should be more than enough to gain relevant feedback.

Design Phase: Evaluation

We plan to evaluate by asking stakeholders for feedback. We currently have two teenage stakeholders who have agreed to participate. We need to be able to give the participants a link in order to have them test the prototype. We have conducted one interview with one of the stakeholders regarding their mis/disinformation skills and knowledge. We hope to be able to finish up these interviews soon. Then we will provide them with the prototype and get their feedback. We are looking to get that done in the next week or so. We are hoping to get feedback on how the participants feel about the prototype in general. We are specifically looking for feedback regarding the usefulness of the prototype, ways the information could be more relevant, and the enjoyment of the prototype. We hope to be able to utilize this feedback in order to improve our prototype. We plan to test the prototype ourselves as well. We may have other members of our class test it, depending on the feedback we receive. We may shift our plan as needed depending on feedback to our prototype. At this moment our projected timeline appears to be operational.

4 Discussion

Key Insights

- **User-centered design is crucial for designing effective educational tools**
  Throughout the design process, we learned that designing effective educational tools requires a user-centered approach. By understanding the needs, preferences, and motivations of the target audience, we can create tools that are engaging and effective in achieving their goals. Through user feedback, we were able to develop a game prototype that was more effective in promoting digital literacy skills among teenagers.

- **Collaboration approaches are essential for addressing complex challenges**
  Another important insight that we gained during the design process was the importance of collaboration and interdisciplinary approaches in addressing complex challenges. Wicked problems, such as the challenge of misinformation and disinformation, require diverse perspectives and skills to address. By collaborating with teammates from different backgrounds and perspectives, we were able to develop a more comprehensive and effective approach to designing a digital literacy game.

- **Flexibility and adaptability are necessary for designing solutions**
  Lastly, we learned that designing solutions requires flexibility and adaptability. Solutions to problems are not static and must be able to evolve and adapt to the problem itself. By designing solutions that are flexible and adaptable, we were able to create a game prototype that could continue to promote digital literacy skills even as the landscape of misinformation and disinformation evolves over time.

Implications-

The project has implications for libraries in that it demonstrates how using gamification could work in terms of mis/disinformation. One of the project's goals was to enable our project to serve as a prototype. We wanted to empower teachers and librarians to utilize our game for their other goals. The prototype could help give libraries an example of how to incorporate mis/disinformation.

The project has implications for the world at large in that it demonstrates the feasibility of this idea. Teachers could construct a game like this with relatively little resources in order to help enhance mis/disinformation education. This prototype could be reconfigured in order to help with any number of subjects. We hope this project could serve a role of demonstrating the feasibility of gamification. We think this demonstration of feasibility is our major implication. The project may also demonstrate the use of things like humor within a learning environment. I think we may have been able to demonstrate the lack of current mis/disinformation education among teens with additional interviews. I think that this is still a reasonable implication, but it could be more provable with additional research.

Limitations and Next Steps

Given the short project timeframe, we were focused on mainly the game design aspect of our solution. To this end, we were able to get valuable feedback from a couple members of our target player audience to get a sense of their feelings about learning about misinformation and what they would like to see in a game targeting this issue. Getting more feedback from a wider range of demographics within our player audience would be ideal. For a more holistic approach, it also would have been helpful to interview education and library professionals to gain insights on how they feel about the format, their thoughts on misinformation as an education, and how they might want to implement the game into a curriculum.

Given more time, we might have expanded the story to include more scenarios that help develop more skills related to identifying mis- and disinformation. This may include exercises focused on other social media platforms or identifying fake posts and headlines from seemingly reputable sources. Additionally, we would have liked to empower teachers to use Twine themselves so they can either add to our existing game or create stories to complement their own lessons. We would do this by creating a simplified guide to Twine that would allow teachers to get started quickly. The software, while simple, may be confusing for others. This guide would be prototyped and evaluation received so it can best serve its purpose. We may have eventually modified this game to be more usable for a middle school audience. We may try to change the language or other features to best accommodate middle schoolers. We would want to consult with middle school teachers and modify the prototype using their advice. We would also get more feedback from middle schoolers. We may have seeked to create different operational versions of our prototype. These prototypes would engage with other kinds of mis/disinformation. These would be designed with the goal of teachers eventually modifying these versions as they see fit. We may do this by creating a less specific version of our prototype.

5 Conclusion

Baloney Detectives uses the framework of a choose-your-own-adventure game to improve students' digital and information literacy. Through four puzzles related to social media misinformation, we aim to improve teenagers' awareness of misinformation technologies and techniques such as misleading sources and data, false health benefits, and deliberately edited information. One area of improvement would have been introducing a wider variety of misinformation topics that are prevalent on social media and the Internet, such as deepfakes and social media bots, but given the time, resources, and small scope, we think that our project does a suitable job at addressing a wide variety of information needs that teens would gain a strong understanding of through our project.

Our design process was informed by two things. Firstly, we did a wide array of background research surrounding the sophistication and trickery that goes into misinformation, and how teens are the most susceptible group. Additionally, we were able to have the perspectives of real-world teens, whose values, needs, and interests we integrated into our finished product and recorded using a variety of design techniques, including value scenarios, user personas, and affinity diagrams. Each of these experiences was very valuable, as we were able to tailor the game to reflect real-world scenarios as closely as possible.

While this project is slightly small, it is still a great example of how educators and information professionals can equip teens with the skills to identify misinformation online, and we see many ways

that this project can expand to not only include a wider range of misinformation issues, but to feature more tutorials as to how both the game itself and the Twine platform as a whole can be used as a teaching tool to support digital media and information literacy education and turn teens into responsible, skeptical digital citizens.

6 References

Atske, Sara. "Teens, Social Media and Technology 2022." *Pew Research Center*, Pew Research Center, 15 Dec. 2022, https://www.pewresearch.org/internet/2022/08/10/teens-social-media-and-technology-2022/

"Eating Disorder Statistics: General & Diversity Stats." *ANAD*, National Association of Anorexia Nervosa and Associated Disorders, 6 Feb. 2023, https://anad.org/eating-disorders-statistics/

Kemp, Simon. "Digital 2021 April Statshot Report." *DataReportal*, DataReportal, 22 Oct. 2021, https://datareportal.com/reports/digital-2021-april-global-statshot

Lee, Jin Ha, and Chris Coward. *The Euphorigen Investigation*. Center for an Informed Public, University of Washington Information School, 2021. https://ischool.uw.edu/capstone/projects/2021/euphorigen-investigation

Sierra, Wendi. *A Strong Fire*. Fort Worth: Texas Christian University, 2022. https://astrongfire.com/